BRIAN LIU

briliu@mit.edu | brianliu97.github.io | 510-676-1131

EDUCATION

Massachusetts Institute of Technology **September 2021 – Present**
Ph.D. Candidate in Operations Research
Advised by Professor Rahul Mazumder

Cornell University **May 2020**
B.S. in Operations Research | GPA: 4.1
Summa Cum Laude | Omega Rho Honor Society

RESEARCH INTERESTS

Topics | Machine learning, operations research, interpretability, discrete and combinatorial optimization, statistics
Applications | Digital platforms, healthcare operations, clinical machine learning, public health, public policy

SELECTED AWARDS AND HONORS

1. 2026 American Statistical Association Nonparametric Statistics Section Best Student Paper Award Finalist
2. 2025 INFORMS Quality, Statistics, and Reliability Section Best Student Paper Competition 1st Place.
3. 2025 American Statistical Association Statistical Computing Section Best Student Paper Award Winner
4. 2024 INFORMS Data Mining Society Best Student Paper Competition 1st Place.
5. 2025 MIT Health and Life Sciences (HEALS) Collaborative Graduate Fellowship.
   - 32 awarded out of 222 applicants.
6. 2025 ISYE-MS&E-IOE Rising Star.

PUBLICATIONS AND WORKING PAPERS

Refereed Journals

1. **B. Liu**, R. Mazumder, and P. Radchenko. Extracting Interpretable Models from Tree Ensembles: Computational and Statistical Perspectives, 2025, Major Revision in the *Journal of the American Statistical Association*, revision submitted.
   - 2026 American Statistical Association Nonparametric Statistics Section Best Student Paper Award Finalist.

2. **B. Liu** and R. Mazumder. Locally Transparent Rule Sets for Interpretable Machine Learning, 2025, R&R in *Operations Research*.
   - 2025 INFORMS Quality, Statistics, and Reliability Section Best Student Paper Competition 1st Place.

3. **B. Liu** and R. Mazumder. Randomization Can Reduce Both Bias and Variance: A Case Study in Random Forests, *Journal of Machine Learning Research*, 2025.

4. **B. Liu***, Y. Zhang*, S. Henderson, D. Shmoys, P. Frazier. Modeling the Risk of In-Person Instruction During the COVID-19 Pandemic. *INFORMS Journal on Applied Analytics*, 2024.

5. P. Frazier, J. M. Cashore, N. Duan, S. Henderson, A. Janmohamed, **B. Liu**, D. Shmoys, J. Wan, Y. Zhang. Modeling for COVID-19 College Reopening Decisions: Cornell, A Case Study. *Proceedings of the National Academy of Sciences*, 2022.

Refereed Conferences:

6. **B. Liu** and R. Mazumder. Moss: Multi-Objective Optimization for Stable Rule Sets. *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (KDD research track), 2025.
   - 19% acceptance rate.
   - 2024 INFORMS Data Mining Society Best Student Paper Competition 1st Place.

7. **B. Liu** and R. Mazumder. Fast: An Optimization Framework for Fast Additive Segmentation. *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (KDD research track), 2024.
   - 20% acceptance rate.
   - 2025 American Statistical Association Statistical Computing Student Paper Competition Winner.

8. **B. Liu** and R. Mazumder. Fire: An Optimization Framework for Fast Interpretable Rule Extraction. *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (KDD research track), 2023.
    - 22% acceptance rate.

9. **B. Liu** and R. Mazumder. ForestPrune: Compact Depth-Pruned Tree Ensembles. *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics* (AISTATS), 2023.
    - 29% acceptance rate.

10. **B. Liu**, M. Xie, and M. Udell. ControlBurn: Feature Selection by Sparse Forests. *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (KDD research track), 2021.
    - 15% acceptance rate.

Refereed Workshops:

11. **B. Liu** and R. Mazumder. TreePrompt: Distilling Boosted Tree Ensembles for In-Context Learning in Large Language Models, *The First Structured Knowledge for Large Language Models Workshop at KDD*, 2025.

Technical Reports:

12. **B. Liu** and M. Udell. Impact of accuracy on model interpretations, 2020, arxiv.org/abs/2011.09903.

TEACHING

Teaching Assistant (MIT) **Spring 2025**
15.071 Analytics Edge (MBA)

Teaching Assistant (MIT) **Fall 2024**
15.072 Advanced Analytics Edge (Graduate Master of Business Analytics)

Teaching Assistant (MIT) **Spring 2024**
15.075 Statistical Thinking and Data Analysis (Undergraduate)

Teaching Assistant (MIT) **Fall 2023**
15.072 Advanced Analytics Edge (Graduate Master of Business Analytics)

Teaching Assistant (MIT) **Fall 2023**
15.071 Analytics Edge (MBA)

Teaching Assistant (MIT) **Summer 2023**
15.067 Engineering Statistics and Data Science (Graduate Master and MBA)

Teaching Assistant (Cornell) **Spring 2020**
ORIE 4740: Introduction to Statistical Learning.

Teaching Assistant (Cornell) **Fall 2018**
ORIE 3300: Optimization I

ADDITIONAL AWARDS

1. KDD 2025 Student Travel Award
2. Invited to 2025 Tippie FutureBAProf Workshop

INDUSTRY EXPERIENCE AND COLLABORATIONS

Graduate Student Researcher **October 2023 – Present**
SilverCloud Health | Boston, MA
Healthcare machine learning project to predict telehealth treatment outcomes for patients with anxiety and depression.

Graduate Student Researcher **July 2022 – May 2024**
Takeda | Cambridge, MA

Clinical machine learning project to improve eosinophilic esophagitis (EoE) diagnosis. Built survival models to improve the timely diagnosis of EoE and machine learning models to improve differential diagnosis.

Graduate Student Researcher **September 2021 – February 2022**
Hartford Healthcare | Hartford, CT
Analyzed patient and provider data to determine the impact of socio-economic factors on disparities in diabetes management.

Data and Applied Scientist **July 2020 – July 2021**
Microsoft (Bing Ads) | Bellevue, WA
Built and maintained machine learning models to forecast advertising account revenue and churn. Analyzed Bing demographic and interest data to optimize audience targeting for online search ads.

Data and Applied Science Intern **Summer 2019**
Microsoft (Devices) | Redmond, WA
Designed an inventory management plan to optimize the safety stock of spare parts in the supply chain.

Data Science Intern **Summer 2018 & Summer 2017**
Tesla (Global Service Operations) | Fremont, CA

TALKS

ACM SIGKDD International Conference on Knowledge Discovery and Data Mining **August 2025**
Multi-Objective Optimization for Stable Rule Sets

Joint Statistical Meeting **August 2025**
Additive Models for Transparent Machine Learning

MIT Industrial Liaison Program Webinar: Unlocking the Value of Employee Wellness **July 2025**
Explainable AI for Digital Mental Health

MIT Sloan Health Systems Initiative Annual Workshop **October 2024**
Interpretable Machine Learning Methods for Predicting Telemental Health Outcomes

INFORMS Annual Meeting **October 2024**
An Optimization Framework for Fast Additive Segmentation in Transparent ML

ACM SIGKDD International Conference on Knowledge Discovery and Data Mining **August 2024**
An Optimization Framework for Fast Additive Segmentation in Transparent ML

International Symposium on Mathematical Programming **July 2024**
An Optimization Framework for Fast Additive Segmentation in Transparent ML

US Census Bureau Center for Statistical Research and Methodology **July 2024**
Making Tree Ensembles Interpretable

ACM SIGKDD International Conference on Knowledge Discovery and Data Mining **August 2023**
Fast Interpretable Rule Extraction

INFORMS Annual Meeting **October 2022**
Depth-Pruning Tree Ensembles

ACM SIGKDD International Conference on Knowledge Discovery and Data Mining **August 2021**
Feature Selection with Sparse Forests

OTHER RESEARCH

Cornell Pandemic Modeling Group **March 2020 – December 2022**
Modeled classroom transmission using Monte Carlo simulation to assess student and instructor risk. Analyzed the impact of university reopening, weather, and compliance to shutdowns on pandemic spread. Presented various data analyses and visualizations to Cornell senior leadership to support decisions on COVID-19 interventions. (https://datasciencecenter.cornell.edu/covid-19-modeling/)

Cornell Pandemic Reopening Planning **March 2020 – September 2020**

Modeled student movement and behavior on campus using mobility and university internal data. Leveraged models to identify congested sidewalks and bus routes and predict potential hotspots for transmission. Tested various de-densification strategies using agent-based simulations and submitted the most effective methods to the provost. Assisted scheduling socially distanced classrooms. (https://www.orie.cornell.edu/spotlights/unsung-engineering-behind-cornells-fall-2020-schedule)

Public Transit Route Optimization **August 2019 – March 2020**

Developed predictive models using farebox and vehicle location data to estimate rider demand on bus routes in Tompkins County. Used demand estimates to redesign the county's second busiest bus route to improve efficiency. Changes were implemented by Tompkins County Area Transit (TCAT) in Fall 2020.

SERVICE

ORC IAP Seminar Coordinator	**Spring 2024**
ORC Student Seminar Coordinator	**Spring 2024**
ORC Student Seminar Coordinator	**Fall 2023**
Reviewer	**Journal of the Royal Statistical Society, Series B**
Reviewer	**Journal of Machine Learning Research**
Reviewer	**ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)**
Reviewer	**International Conference on Artificial Intelligence and Statistics (AISTATS)**

TECHNICAL SKILLS

Python, Julia, R, SQL, Spark, Databricks, AMPL, JuMP, Gurobi, Tableau, LaTeX